

Quiz Week 8

Groups



The order of an element in a cyclic group

1. Let $G = \langle g \rangle$ be a cyclic group of order 20. Find the order of the following elements of G :

(a) $\text{o}(g^2)$.

(b) $\text{o}(g^5)$.

Solution: We apply Lemma 3.31 from the notes, which says that if $d|n$, then $\text{o}(g^d) = n/d$.

(a) $\text{o}(g^2) = \frac{20}{2} = 10$.

(b) $\text{o}(g^5) = \frac{20}{5} = 4$.

2. In the additive cyclic group $\mathbb{Z}_{15}$, find the order of:

(a) $\bar{3}$,

(b) $\bar{2}$,

(c) $\bar{10}$.

Solution: Let us begin by noticing that $\mathbb{Z}_1 5 = \langle \bar{1} \rangle$, and $\bar{1} 5 = 15\bar{1}$.

(a) It is clear that $\bar{3} = 3\bar{1}$. Since $3|15$, we can apply the additive version of Lemma 3.31 (from our notes) to get that the order of $\bar{3}$ is $\frac{15}{3} = 5$. In this case, we can write down all the elements of $\langle \bar{3} \rangle$:

$$\langle \bar{3} \rangle = \{\bar{0}, \bar{3}, \bar{6}, \bar{9}, \bar{12}\}.$$

(b) In this case, we have that $\text{gcd}(2, 15) = 1$. This tells us that $\bar{2}$ is another generator of $\mathbb{Z}_{15}$ (check Theorem 3.28). Thus the order of $\bar{2}$ is 15.

(c) Here, we first apply the additive version of Corollary 3.32, which says that

$$\langle \bar{10} \rangle = \langle \overline{\text{gcd}(10, 15)} \rangle = \langle \bar{5} \rangle.$$

Next, we find the order of $\bar{5}$ using the additive version of Lemma 3.31: the order of $\bar{5}$ is now $\frac{15}{5} = 3$. Thus, the order of $\bar{10}$ is 3. In fact:

$$\langle \bar{10} \rangle = \{\bar{0}, \bar{10}, \bar{20}\} = \{\bar{0}, \bar{5}, \bar{10}\} = \langle \bar{5} \rangle.$$